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| | BANNARI AMMAN INSTITUTE OF TECHNOLOGY<br>(An Autonomous Institution Affiliated to Anna University, Chennai)<br>SATHYAMANGALAM - 638 401 |
| | Module Test - II - May - 2026<br>II Semester |

## 22GE002 & COMPUTATIONAL PROBLEM SOLVING

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| 1 | (i) | A school stores student data in a flat-file database. The system allows only basic search operations and does not support advanced queries. Duplicate records are often entered without any validation.<br>Examine the characteristics of the system and explain how they affect data management.<br>(3 Marks - [An/F, 2]) |
| 2 | (ii) | A small library maintains a list of books with details like title, author, and price in a single file. The system handles only a small amount of data and does not require linking between different records.<br>Explain why a flat-file database is an appropriate choice for this scenario.<br>(2 Marks - [U/F, 1]) |
| 3 | (i) | Determine the appropriate cloud storage type used in each application with suitable justification.<br>A startup company stores its website data and backups using services like AWS and Google Cloud to reduce infrastructure cost and improve scalability.<br>A bank maintains its customer transaction data in a dedicated storage environment that is not shared with other organizations due to strict security requirements.<br>A multinational company stores sensitive employee records in a private environment while keeping general application data in a public cloud for better scalability.<br>(3 Marks - [U/C, 2]) |
| 4 | (ii) | A media company needs cloud storage to manage large video files and backups, while another organization requires secure cloud storage with collaboration and compliance features for enterprise use.<br>Determine the suitable cloud storage providers for these requirements with justification.<br>(2 Marks - [U/C, 1]) |
| 5 | (i) | An organization manages different types of data for its operations. Sales records are stored in tables, web data is collected in XML format, and emails and images are stored separately. These data types are used for reporting, communication, and content management.<br>Classify the data given in the scenario into appropriate type with justification<br>(3 Marks - [U/P, 1]) |
| 6 | (ii) | A small tuition center maintains student details like name, phone number, and marks in a single Excel sheet. There are no separate tables or relationships.<br>Identify the type of database used and mention one key characteristic.<br>(2 Marks - [U/P, 1]) |

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| 7 | (i) | An employee stores important project files in a cloud storage service and accesses them from different locations using the internet. The employee logs into the cloud platform, retrieves the required file, and uses it for further processing. The system ensures secure and reliable access to stored data.<br>Write the steps involved in reading data from cloud storage in the above scenario.<br>(3 Marks - [U/C, 2]) |
| 8 | (ii) | A company uses cloud storage to access data from different devices, increase storage capacity when needed, and recover important files during system failures.<br>Outline any two benefits of cloud storage highlighted in the above scenario with suitable justification.<br>(2 Marks - [U/P, 1]) |
| 9 | (i) | A college maintains student details in one table and department information with HOD details in another table. The administration needs to combine and analyze data from both tables to manage academic records effectively. |

Student Table

| StudentID | Name | Dept | Address |
|---|---|---|---|
| 1 | Ravi | CSE | Chennai |
| 2 | Meena | ECE | Coimbatore |
| 3 | Arjun | CSE | Madurai |
| 4 | Kiran | IT | Trichy |
| 5 | Divya | ECE | Salem |
| 6 | Suresh | MECH | Erode |

Department Table

| DeptID | DeptName | HODName |
|---|---|---|
| D1 | CSE | Dr. Kumar |
| D2 | IT | Dr. Aravind |
| D3 | MECH | Dr. Suresh |
| D4 | EEE | Dr. Priya |
| D5 | CIVIL | Dr. Ramesh |

Using the given Students and Departments tables, determine the output of the following relational algebra operations:

a) Retrieve the HOD name of the MECH department by applying a selection operation on the Departments table.

b) Display only the names of students from the Students table using a projection operation.

c) Using the DeptName values from both tables, determine the resulting tables for the union, intersection, and difference operations.

(5 Marks - [U/C, 3])

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| 10 | (i) | Consider the below employee table to answer the following questions.<br><table><tr><td>Eid</td><td>Ename</td><td>Dept</td><td>Email_Id</td><td>Phone</td><td>Salary</td></tr><tr><td>101</td><td>Adams</td><td>Research</td><td>adams@gmail.com</td><td>11111</td><td>50000.00</td></tr><tr><td>102</td><td>Allen</td><td>Sales</td><td>allen@gmail.com</td><td>22222</td><td>15000.50</td></tr><tr><td>103</td><td>Blake</td><td>Sales</td><td>blake@gmail.com</td><td>33333</td><td>18000.00</td></tr><tr><td>104</td><td>Clark</td><td>Accounting</td><td>clark@gmail.com</td><td>44444</td><td>20000.75</td></tr><tr><td>105</td><td>Ford</td><td>Research</td><td>ford@gmail.com</td><td>55555</td><td>36000.25</td></tr><tr><td>106</td><td>James</td><td>Sales</td><td>james@gmail.com</td><td>66666</td><td>12000.50</td></tr></table>a) Drop the column phone number from the Employee table and display the output.<br><br>b) Modify the datatype of salary column to integer in the employee table and display the output.<br><br>(5 Marks - [U/C, 3]) |
| 11 | (i) | A hospital database manages patient, doctor, and insurance information using various constraints. Each patient is assigned a unique ID to avoid duplication. The system enforces valid ranges for attributes such as age and restricts deletion of doctor records that are linked to patients. It also applies specific rules to determine insurance eligibility based on predefined conditions.<br>Illustrate how different types of data integrity are applied in this system and discuss their impact on data accuracy.<br><br>(5 Marks - [U/P, 3]) |
| 12 | (i) | A college database administrator needs to manage the structure of a student database system. The administrator creates new tables for storing student details, modifies existing table structures to add new attributes, removes unnecessary tables, and renames tables when required.<br><br>Illustrate how DDL (Data Definition Language) commands are used in the above scenario and discuss the role of commands such as CREATE, ALTER, DROP, and RENAME in database management.<br><br>(5 Marks - [U/C, 3]) |
| 13 | (i) | An organization uses different servers to support its operations. Employee's access websites through a system that delivers web pages, store and retrieve data from a centralized system, share files within the office network, send and receive emails, and resolve domain names while browsing. Some systems also act as intermediaries to filter content and improve security.<br><br>Classify any five types of servers used in the given scenario and explain the function of each in supporting organizational activities.<br><br>(5 Marks - [U/C, 3]) |

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| 14 | (i) | Consider the below student table to answer the following questions.<br>Student |

| Id | Name | Address | Age | Department | Marks |
|---|---|---|---|---|---|
| 1 | Raju | Chennai | 25 | Arts | 92 |
| 2 | Ramesh | Kashmir | 25 | Engish | 89 |
| 3 | Mani | Delhi | 26 | Arts | 81 |
| 4 | John | Delhi | 26 | English | 85 |

a) Insert two rows into the student table. After entering value, the new table should appear as like the below table by performing SQL Query.

| Id | Name | Address | Age | Department | Marks |
|---|---|---|---|---|---|
| 1 | Raju | Chennai | 25 | Arts | 92 |
| 2 | Ramesh | Kashmir | 25 | Engish | 89 |
| 3 | Mani | Delhi | 26 | Arts | 81 |
| 4 | John | Delhi | 26 | English | 85 |
| 5 | Thiyagarajan | Vizag | 27 | Arts | 91 |
| 6 | Surendran | Pune | 28 | History | 96 |

b) Write an SQL query to perform the following operations,

i) Update the address of student named Mani to Mumbai

ii) Delete the student from the student table whose id is 5.

(5 Marks - [U/C, 3])

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| 15 | (i) | An organization plans to upgrade its storage infrastructure to handle increasing data demands. It needs a solution that provides both high performance and protection against data loss. Different RAID configurations are being evaluated to meet these requirements. The goal is to choose the most suitable RAID level based on speed and reliability.<br><br>Compare RAID 0, RAID 1, RAID 5, and RAID 10 in terms of performance and data reliability. Also, Evaluate the RAID levels and determine which configuration provides the highest data reliability.<br><br>(5 Marks - [U/C, 3]) |
| 16 | (i) | A banking system stores confidential customer account details in a database. Managers are given permission to view and update customer information for banking operations. Unauthorized employees should not be allowed to modify sensitive records. The database administrator manages these permissions to maintain data security and controlled access. Illustrate how GRANT and REVOKE commands can be applied to control user privileges in this system with an example query.<br><br>(5 Marks - [U/C, 3]) |
| 17 | (i) | A user sends a video file from a laptop to a smart TV through a network. The data travels through a fiber optic cable, and both devices follow a common set of communication rules to successfully exchange information.<br><br>Identify the components of the data communication system involved in this scenario and explain their roles.<br><br>(3 Marks - [U/C, 2]) |

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| 18 | (ii) | A university connects multiple departments using a wired network to enable communication between students and faculty. During file transfer, a student sends a project file intended for a professor, but the file is incorrectly received by another student in a different department.<br><br>Identify the issue in the communication system and indicate the characteristic of data communication that has been violated in this scenario.<br><br>(2 Marks - [U/C, 1]) |
| 19 | (i) | A company assigns subnets A, B, C, and D from the network 192.168.12.0. Each subnet must support communication within its range without overlap.<br><br>Identify the network addresses of all four subnets.<br><br>(3 Marks - [U/C, 1]) |
| 20 | (ii) | In a small office network, the administrator manually assigns IP addresses to printers and servers, while user devices automatically receive IP addresses from a central system.<br><br>Identify which devices use static IP assignment and which devices use dynamic IP assignment.<br><br>(2 Marks - [U/P, 1]) |
| 21 | (i) | In a data communication system, different types of data flow are used to control the direction of data transmission between devices. Consider a scenario where two devices are connected in such a way that both can transmit and receive data at the same time without waiting for each other.<br><br>Identify the type of data flow used in this system and differentiate it clearly from the other types of data flow used in data communication.<br><br>(3 Marks - [U/C, 2]) |
| 22 | (ii) | A company uses a network to transfer important documents between two offices. During transmission, some parts of the document are altered, resulting in incorrect information at the receiving end.<br><br>Identify the issue in the communication system and indicate the characteristic of data communication that has been violated in this scenario.<br><br>(2 Marks - [U/P, 1]) |
| 23 | (i) | A network administrator divides the network 192.168.12.0 into four equal subnets (A, B, C, and D) to allocate IP addresses efficiently across different departments. Each subnet has a defined network address and broadcast address.<br>Based on the given subnetting, identify the network address of subnet B, the broadcast address of subnet C, and determine the subnet in which the IP address 192.168.12.200 belongs.<br><br>(3 Marks - [U/P, 2]) |
| 24 | (ii) | In an organization, network devices that require постоян connectivity are configured manually, while temporary devices are assigned IP addresses by a network service.<br><br>Indicate the type of IP assignment used for permanent devices and temporary devices.<br><br>(2 Marks - [U/C, 1]) |

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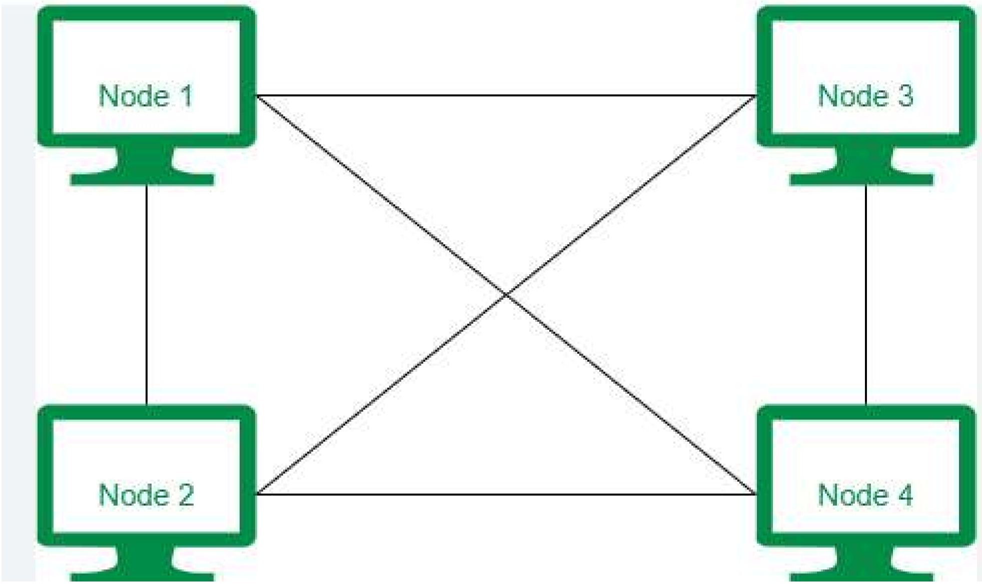
| 25 | (i) | A network diagram shows four nodes (Node 1, Node 2, Node 3, Node 4) where each node is directly connected to every other node, forming multiple communication links between devices.<br>Analyze how data communication occurs in this network structure and explain how the connectivity between nodes affects reliability and data transmission. Also evaluate how the network behaves if one connection fails.<br><br><br>Figure:Mesh Topology<br><br>(5 Marks - [U/P, 3]) |
| 26 | (i) | A university is designing a campus network that connects multiple buildings such as academic blocks, administration offices, and hostels. The network is structured using core, distribution, and access layers to ensure efficient communication and scalability. During operation, issues such as data congestion, uneven traffic distribution, and slow communication between departments are observed.<br><br>Explain how the core, distribution, and access layers function in this campus network. Also, describe how data flows between different LANs and discuss the role of load balancing in improving network performance.<br><br>(5 Marks - [U/C, 3]) |
| 27 | (i) | A company plans to design a network for its office where communication between devices should be easy to manage, and adding or removing devices should not affect the overall network structure.<br>Analyze the requirement and explain which network topology is most suitable. Also describe how this topology supports communication and network management.<br><br>(5 Marks - [U/P, 3]) |

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| 28 | (i) | A student attempts to access the website www.campus.edu from a campus network. The system must resolve the domain name into an IP address using the DNS hierarchy. During this process, multiple DNS servers are involved before the final IP address is obtained. Explain the step-by-step process of domain name resolution, clearly describing the role of root servers, TLD servers, and authoritative servers. Also, distinguish between forward and reverse domain name services.<br><br>(5 Marks - [U/P, 3]) |
| 29 | (i) | A network administrator is designing networks for different organizations such as a multinational company, a large organization, and a small business. The administrator must assign suitable IP address classes based on the size and requirement of each network.<br><br>Analyze how IPv4 address classification supports this requirement and explain how different classes are suitable for networks of varying sizes and purposes.<br><br>(5 Marks - [U/C, 3]) |
| 30 | (i) | A university is designing a campus network to connect multiple departments located in different buildings. The network must support high-speed data transfer, apply routing policies between departments, and provide reliable access to end-user devices like computers and printers.<br><br>Analyze how the core layer, distribution layer, and access layer contribute to the efficient configuration and management of the campus network.<br><br>(5 Marks - [U/C, 3]) |
| 31 | (i) | A company is assigned the IP address 192.168.10.25 with subnet mask 255.255.255.0. Identify the network address and host address, and indicate how the given IP address is divided into network and host portions using the subnet mask.<br><br>(5 Marks - [U/P, 3]) |

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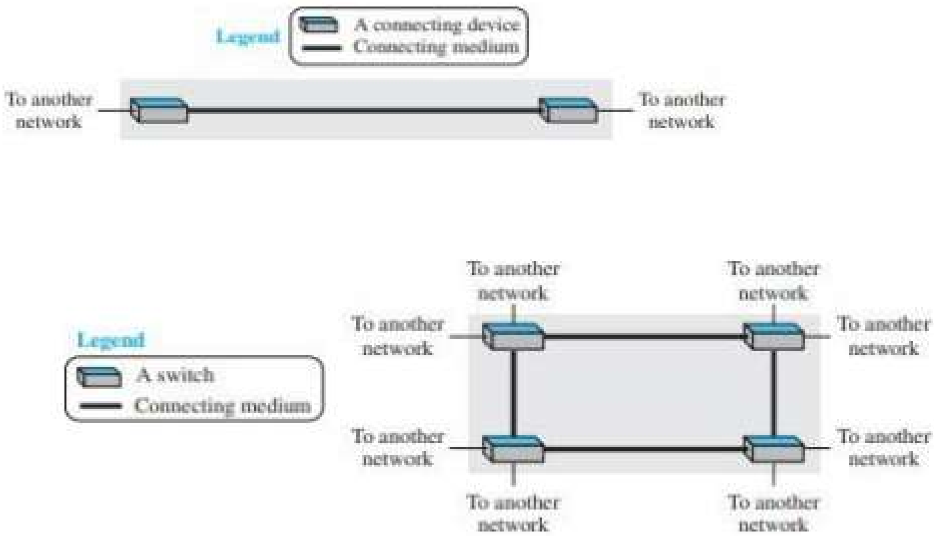
| 32 | (i) | A multinational company operates offices in different cities and countries. The head office needs to communicate with two remote branches using a dedicated communication link for secure data transfer. At the same time, multiple other branch offices are interconnected through a shared network infrastructure that allows data exchange between many locations.<br>Based on this scenario,<br>identify and differentiate the types of WAN connections used for:<br>Communication between the head office and two specific branches<br>Communication among multiple branch offices<br>Also, justify which WAN type is more suitable for global communication backbone and why.<br><br><br>Figure : Types of WAN<br>(5 Marks - [U/C, 3]) |